Does Technology Drive History?

The Dilemma of Technological Determinism

edited by Merritt Roe Smith and Leo Marx

ization that as a national culture we are not Daniel Boone, or Bonnie and Clyde, or even the crew of the Starship *Enterprise.* We have more nearly resembled a latter-day Walter Mitty, jarred from our day-dreaming to discover that the landscapes we drive through, the fuel we consume, and the air we pollute do not get left behind as we drift on to yet another "final frontier."[12] Our elusive destination may turn out to be nearer than we thought. In the 1960s, Marshall McLuhan celebrated the "global village"; we have since discovered that all of us who inhibit that village are "downwinders," consuming the blessings and the curses of technology with every breath. Our new frontiers are not geographic.

What would an update of "Science on the March" look like in 2002? Historians are reluctant to describe the future, but I am happy to propose a utopian fantasy to help fill the void that will surely follow the demise of technological determinism: a terrain characterized, not by the headlong commuter-rush toward distant horizons of progress, but by a homegrown version of *The Day the Earth Stood Still.* For a brief period, everything stops. Everyone gets out of the car of the future, walks about, talks with everyone else about what needs to be done. They begin to devise their own maps for roads not taken. They agree that social expertise requires collective effort and should no longer be confused with technical expertise. They redefine national security, dismantling Stealth bombers and SDI laser satellite systems and converting them into prenatal care, universal health coverage, safe energy, clean air and water, reliable mass transit, and the elimination of hunger. They elect to replace pesticides with Integrated Pest Management techniques. They begin a conversion project to remove the great dams from Western rivers and to minimize the environmental effects of manufacturing. Defense lobbyists, sensing that their jobs are now superfluous, choose to become historians, and set about recording the stories unfolding all around them, in the national interest.

12. Langdon Winner, *The Whale and the Reactor: A Search for Limits in an Age of High Technology* (University of Chicago Press, 1986), p. 10.

Do Machines Make History?

Robert L. Heilbroner

Among the scholars represented in this anthology, economic historian Robert Heilbroner comes closest to embracing technological determinism. But he does so with carefully stated qualifications. Heilbroner's now-classic 1967 essay seeks to specify the extent to which technology determines "the nature of the socioeconomic order." He does so by showing that technological change follows a roughly ordered sequence of development and that it "imposes certain social and political characteristics upon the society in which it is found." Yet, rather than take a strong stand for technological determinism, Heilbroner opts for a "soft" version, because he sees a complex historical scenario in which technology, while acting on society, also reflects the influence of socioeconomic forces on its development. Ultimately he views technology as a strong "mediating factor" rather than as the *determining influence on history—a point he reiterates and expands upon in the retrospective essay that follows this one.*

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"The hand-mill gives you society with the feudal lord; the steam-mill, society with the industrial capitalist."

—*Marx,* The Poverty of Philosophy

That machines make history in some sense—that the level of technology has a direct bearing on the human drama—is of course obvious. That they do not make all of history, however that word be defined, is equally clear. The challenge, then, is to see if one can say something systematic about the matter, to see whether one can order the problem so that it becomes intellectually manageable.

To do so calls at the very beginning for a careful specification of our task. There are a number of important ways in which machines make history that will not concern us here. For example, one can study the impact of technology on the *political* course of history, evidenced most strikingly by the central role played by the technology of war. Or one can study the effect of machines on the *social* attitudes that underlie historical evolution: one thinks of the effect of radio or television on political behavior. Or one can study technology as one of the factors shaping the changeful content of life from one epoch to another: when we speak of "life" in the Middle Ages or today we define an existence much of whose texture and substance is intimately connected with the prevailing technological order.

None of these problems will form the focus of this essay. Instead, I propose to examine the impact of technology on history in another area—an area defined by the famous quotation from Marx that appears above. The question we are interested in, then, concerns the effect of technology in determining the nature of the *socioeconomic order.* In its simplest terms the question is: Did medieval technology bring about feudalism? Is industrial technology the necessary and sufficient condition for capitalism? Or, by extension, will the technology of the computer and the atom constitute the ineluctable cause of a new social order?

Even in this restricted sense, our inquiry promises to be broad and sprawling. Hence, I shall not try to attack it head-on, but to examine it in two stages:

1. If we make the assumption that the hand-mill does "give" us feudalism and the steam-mill capitalism, this places technological change in the position of a prime mover of social history. Can we then explain the "laws of motion" of technology itself? Or, to put the question less grandly, can we explain why technology evolves in the sequence it does?

2. Again, taking the Marxian paradigm at face value, exactly what do we mean when we assert that the hand-mill "gives us" society with the feudal lord? Precisely how does the mode of production affect the superstructure of social relationships?

These questions will enable us to test the empirical content—or at least to see if there *is* an empirical content—in the idea of technological determinism. I do not think it will come as a surprise if I announce now that we will find *some* content, and a great deal of missing evidence, in our investigation. What will remain then will be to see if we can place the salvageable elements of the theory in historical perspective—to see, in a word, if we can explain technological determinism historically as well as explain history by technological determinism.

I

We begin with a very difficult question hardly rendered easier by the fact that there exist, to the best of my knowledge, no empirical studies on which to base our speculations. It is the question of whether there is a fixed sequence to technological development and therefore a necessitous path over which technologically developing societies must travel.

I believe there is such a sequence—that the steam-mill follows the hand-mill not by chance but because it is the next "stage" in a technical conquest of nature that follows one and only one grand avenue of advance. To put it differently, I believe that it is impossible to proceed to the age of the steam-mill until one has passed through the age of the hand-mill, and that in turn one cannot move to the age of the hydroelectric plant before one has mastered the steam-mill, nor to the nuclear power age until one has lived through that of electricity.

Before I attempt to justify so sweeping an assertion, let me make a few reservations. To begin with, I am fully conscious that not all societies are interested in developing a technology of production or in channeling to it the same quota of social energy. I am very much aware of the different pressures that different societies exert on the direction in which technology unfolds. Lastly, I am not unmindful of the difference between the discovery of a given machine and its application as a technology—for example, the invention of a steam engine (the aeolipile) by Hero of Alexandria long before its incorporation into a steam-mill. All these problems, to which we will return in our last section, refer, however, to the way in which technology makes its peace with the social, political, and economic institutions of the society in which it appears. They do not directly affect the contention that there exists a determinate sequence of productive technology for those societies that are interested in originating and applying such a technology.

What evidence do we have for such a view? I would put forward three suggestive pieces of evidence:

The Simultaneity of Invention

The phenomenon of simultaneous discovery is well known.[1] From our view, it argues that the process of discovery takes place along a well-defined frontier of knowledge rather than in grab-bag fashion. Admittedly, the concept of "simultaneity" is impressionistic,[2] but the related phenomenon of technological "clustering" again suggests that technical evolution follows a sequential and determinate rather than random course.[3]

The Absence of Technological Leaps

All inventions and innovations, by definition, represent an advance of the art beyond existing base lines. Yet, most advances, particularly in retrospect, appear essentially incremental, evolutionary. If nature

1. See Robert K. Merton, "Singletons and Multiples in Scientific Discovery: A Chapter in the Sociology of Science," *Proceedings of the American Philosophical Society* 105 (October 1961): 470–486.

2. See John Jewkes, David Sawers, and Richard Stillerman, *The Sources of Invention* (New York, 1960 [paperback edition]), p. 227, for a skeptical view.

3. "One can count 21 basically different means of flying, at least eight basic methods of geophysical prospecting; four ways to make uranium explosive; . . . 20 or 30 ways to control birth. . . . If each of these separate inventions were autonomous, i.e., without cause, how could one account for their arriving in these functional groups?" S. C. Gilfillan, "Social Implications of Technological Advance," *Current Sociology* 1 (1952): 197. See also Jacob Schmookler, "Economic Sources of Inventive Activity," *Journal of Economic History* (March 1962): 1–20; Richard Nelson, "The Economics of Invention: A Survey of the Literature," *Journal of Business* 32 (April 1959): 101–119.

makes no sudden leaps, neither, it would appear, does technology. To make my point by exaggeration, we do not find experiments in electricity in the year 1500, or attempts to extract power from the atom in the year 1700. On the whole, the development of the technology of production presents a fairly smooth and continuous profile rather than one of jagged peaks and discontinuities.

The Predictability of Technology

There is a long history of technological prediction, some of it ludicrous and some not.[4] What is interesting is that the development of technical progress has always seemed *intrinsically* predictable. This does not mean that we can lay down future timetables of technical discovery, nor does it rule out the possibility of surprises. Yet I venture to state that many scientists would be willing to make *general* predictions as to the nature of technological capability 25 or even 50 years ahead. This, too, suggests that technology follows a developmental sequence rather than arriving in a more chancy fashion.

I am aware, needless to say, that these bits of evidence do not constitute anything like a "proof" of my hypothesis. At best they establish the grounds on which a *prima facie* case of plausibility may be rested. But I should like now to strengthen these grounds by suggesting two deeper-seated reasons why technology *should* display a "structured" history.

The first of these is that a major constraint always operates on the technological capacity of an age, the constraint of its accumulated stock of available knowledge. The application of this knowledge may lag behind its reach; the technology of the hand-mill, for example, was by no means at the frontier of medieval technical knowledge, but technical realization can hardly precede what men generally know (although experiment may incrementally advance both technology and knowledge concurrently). Particularly from the mid-nineteenth century to the present do we sense the loosening constraints on technology stemming from successively yielding barriers of scientific

4. Jewkes et al. (see n. 2) present a catalogue of chastening mistakes (p. 230f.). On the other hand, for a sober predictive effort, see Francis Bellow, "The 1960s: A Forecast of Technology," *Fortune* 59 (January 1959): 74–78; Daniel Bell, "The Study of the Future," *Public Interest* 1 (fall 1965): 119–130. Modern attempts at prediction project likely avenues of scientific advance or technological function rather than the feasibility of specific machines.

knowledge—loosening constraints that result in the successive arrival of the electrical, chemical, aeronautical, electronic, nuclear, and space stages of technology.[5]

The gradual expansion of knowledge is not, however, the only order-bestowing constraint on the development of technology. A second controlling factor is the material competence of the age, its level of technical expertise. To make a steam engine, for example, requires not only some knowledge of the elastic properties of steam but the ability to cast iron cylinders of considerable dimensions with tolerable accuracy. It is one thing to produce a single steam machine as an expensive toy, such as the machine depicted by Hero, and another to produce a machine that will produce power economically and effectively. The difficulties experienced by Watt and Boulton in achieving a fit of piston to cylinder illustrate the problems of creating a technology, in contrast with a single machine.

Yet until a metal-working technology was established—indeed, until an embryonic machine-tool industry had taken root—an industrial technology was impossible to create. Furthermore, the competence required to create such a technology does not reside alone in the ability or inability to make a particular machine (one thinks of Babbage's ill-fated calculator as an example of a machine born too soon), but in the ability of many industries to change their products or processes to "fit" a change in one key product or process.

This necessary requirement of technological congruence[6] gives us an additional cause of sequencing. For the ability of many industries to cooperate in producing the equipment needed for a "higher" stage of technology depends not alone on knowledge or sheer skill but on the division of labor and the specialization of industry. And this in turn hinges to a considerable degree on the sheer size of the stock of capital itself. Thus the slow and painful accumulation of capital,

5. To be sure, the inquiry now regresses one step and forces us to ask whether there are inherent stages for the expansion of knowledge, at least insofar as it applies to nature. This is a very uncertain question. But having already risked so much, I will hazard the suggestion that the roughly parallel sequential development of scientific understanding in those few cultures that have cultivated it (mainly classical Greece, China, the high Arabian culture, and the West since the Renaissance) makes such a hypothesis possible, provided that one looks to broad outlines and not to inner detail.

6. The phrase is Richard LaPiere's in *Social Change* (New York, 1965), p. 263f.

from which springs the gradual diversification of industrial function, becomes an independent regulator of the reach of technical capability.

In making this general case for a determinate pattern of technological evolution—at least insofar as that technology is concerned with production—I do not want to claim too much. I am well aware that reasoning about technical sequences is easily faulted as *post hoc ergo propter hoc*. Hence, let me leave this phase of my inquiry by suggesting no more than that the idea of a roughly ordered progression of productive technology seems logical enough to warrant further empirical investigation. To put it as concretely as possible, I do not think it is just by happenstance that the steam-mill follows, and does not precede, the hand-mill, nor is it mere fantasy in our own day when we speak of the coming of the automatic factory. In the future as in the past, the development of the technology of production seems bounded by the constraints of knowledge and capability and thus, in principle at least, open to prediction as a determinable force of the historical process.

II

The second proposition to be investigated is no less difficult than the first. It relates, we will recall, to the explicit statement that a given technology imposes certain social and political characteristics upon the society in which it is found. Is it true that, as Marx wrote in *The German Ideology*, "A certain mode of production, or industrial stage, is always combined with a certain mode of cooperation, or social stage,"[7] or that, as he put it in the sentence immediately preceding our hand-mill–steam-mill paradigm, "in acquiring new productive forces men change their mode of production, and in changing their mode of production they change their way of living—they change all their social relations"?

As before, we must set aside for the moment certain "cultural" aspects of the question. But if we restrict ourselves to the functional relationships directly connected with the process of production itself, I think we can indeed state that the technology of a society imposes a determinate pattern of social relations on that society.

7. Karl Marx and Friedrich Engels, *The German Ideology* (London, 1942), p. 18.

We can, as a matter of fact, distinguish at least two such modes of influence:

The Composition of the Labor Force

In order to function, a given technology must be attended by a labor force of a particular kind. Thus, the hand-mill (if we may take this as referring to late medieval technology in general) required a work force composed of skilled or semiskilled craftsmen, who were free to practice their occupations at home or in a small atelier, at times and seasons that varied considerably. By way of contrast, the steam-mill—that is, the technology of the nineteenth century—required a work force composed of semiskilled or unskilled operatives who could work only at the factory site and only at the strict time schedule enforced by turning the machinery on or off. Again, the technology of the electronic age has steadily required a higher proportion of skilled attendants; and the coming technology of automation will still further change the needed mix of skills and the locale of work, and may as well drastically lessen the requirements of labor time itself.

The Hierarchical Organization of Work

Different technological apparatuses require not only different labor forces but different orders of supervision and coordination. The internal organization of the eighteenth-century handicraft unit, with its typical man-master relationship, presents a social configuration of a wholly different kind from that of the nineteenth-century factory with its men-manager confrontation, and this in turn differs from the internal social structure of the continuous-flow, semi-automated plant of the present, as the intricacy of the production process increases, a much more complex system of internal controls is required to maintain the system in working order.

Does this add up to the proposition that the steam-mill gives us society with the industrial capitalist? Certainly the class characteristics of a particular society are strongly implied in its functional organization. Yet it would seem wise to be very cautious before relating political effects exclusively to functional economic causes. The Soviet Union, for example, proclaims itself to be a socialist society although its technical base resembles that of old-fashioned capitalism. Had Marx written that the steam-mill gives you society with the industrial *manager*, he would have been closer to the truth.

What is less easy to decide is the degree to which the technological infrastructure is responsible for some of the sociological features of society. Is anomie, for instance, a disease of capitalism or of all industrial societies? Is the organization man a creature of monopoly capital or of all bureaucratic industry wherever found? These questions tempt us to look into the problem of the impact of technology on the existential quality of life, an area we have ruled out of bounds for this paper. Suffice it to say that superficial evidence seems to imply that the similar technologies of Russia and America are indeed giving rise to similar social phenomena of this sort.

As with the first portion of our inquiry, it seems advisable to end this section on a note of caution. There is a danger, in discussing the structure of the labor force or the nature of intrafirm organization, of assigning the sole causal efficacy to the visible presence of machinery and of overlooking the invisible influences of other factors at work. Gilfillan, for instance, writes, "engineers have committed such blunders as saying the typewriter brought women to work in offices, and with the typesetting machine made possible the great modern newspaper, forgetting that in Japan there are women office workers and great modern newspapers getting practically no help from typewriters and typesetting machines."[8] In addition, even where technology seems unquestionably to play the critical role, an independent "social" element unavoidably enters the scene in the *design* of technology, which must take into account such facts as the level of education of the work force or its relative price. In this way the machine will reflect, as much as mold, the social relationships of work.

These caveats urge us to practice what William James called a "soft determinism" with regard to the influence of the machine on social relations. Nevertheless, I would say that our cautions qualify rather than invalidate the thesis that the prevailing level of technology imposes itself powerfully on the structural organization of the productive side of society. A foreknowledge of the shape of the technical core of society 50 years hence may not allow us to describe the political attributes of that society, and may perhaps only hint at its sociological character, but assuredly it presents us with a profile of requirements, both in labor skills and in supervisory needs, that differ considerably from those of today. We cannot say whether the society

8. Gilfillan (see n. 3), p. 202.

of the computer will give us the latter-day capitalist or the commissar, but it seems beyond question that it will give us the technician and the bureaucrat.

III

Frequently, during our efforts thus far to demonstrate what is valid and useful in the concept of technological determinism, we have been forced to defer certain aspects of the problem until later. It is time now to turn up the rug and to examine what has been swept under it. Let us try to systematize our qualifications and objections to the basic Marxian paradigm:

Technological Progress Is Itself a Social Activity

A theory of technological determinism must contend with the fact that the very activity of invention and innovation is an attribute of some societies and not of others. The Kalahari bushmen or the tribesmen of New Guinea, for instance, have persisted in a neolithic technology to the present day; the Arabs reached a high degree of technical proficiency in the past and have since suffered a decline; the classical Chinese developed technical expertise in some fields while unaccountably neglecting it in the area of production. What factors serve to encourage or discourage this technical thrust is a problem about which we know extremely little at the present moment.[9]

The Course of Technological Advance Is Responsive to Social Direction

Whether technology advances in the area of war, the arts, agriculture, or industry depends in part on the rewards, inducements, and incentives offered by society. In this way the direction of technological advance is partially the result of social policy. For example, the system of interchangeable parts, first introduced into France and then independently into England, failed to take root in either country for lack of government interest or market stimulus. Its success in America is attributable mainly to government support and to its appeal in a

9. An interesting attempt to find a line of social causation is found in E. Hagen, *The Theory of Social Change* (Homewood, Ill., 1962).

society without guild traditions and with high labor costs.[10] The general *level* of technology may follow an independently determined sequential path, but its areas of application certainly reflect social influences.

Technological Change Must be Compatible with Existing Social Conditions

An advance in technology not only must be congruent with the surrounding technology but must also be compatible with the existing economic and other institutions of society. For example, labor-saving machinery will not find ready acceptance in a society where labor is abundant and cheap as a factor of production. Nor would a mass-production technique recommend itself to a society that did not have a mass market. Indeed, the presence of slave labor seems generally to inhibit the use of machinery and the presence of expensive labor to accelerate it.[11]

These reflections on the social forces bearing on technical progress tempt us to throw aside the whole notion of technological determinism as false or misleading.[12] Yet, to relegate technology from an undeserved position of *primum mobile* in history to that of a mediating factor, both acted upon by and acting on the body of society, is not to write off its influence but only to specify its mode of operation with greater precision. Similarly, to admit we understand very little of the cultural factors that give rise to technology does not depreciate its role but focuses our attention on that period of history when technology is clearly a major historical force, namely Western society since 1700.

IV

What is the mediating role played by technology within modern Western society? When we ask this much more modest question, the interaction of society and technology begins to clarify itself for us.

10. See K. R. Gilbert, "Machine-Tools," in *A History of Technology*, ed. C. Singer et al. (Oxford, 1958), IV, chapter xiv.

11. See LaPiere (n. 6), p. 284; also H. J. Habbakuk, *British and American Technology in the 19th Century* (Cambridge, 1962), passim.

12. As, for example, in A. Hansen, "The Technological Determination of History," *Quarterly Journal of Economics* (1921): 76–83.

The Rise of Capitalism Provided a Major Stimulus for the Development of a Technology of Production

Not until the emergence of a market system organized around the principle of private property did there also emerge an institution capable of systematically guiding the inventive and innovative abilities of society to the problem of facilitating production. Hence the environment of the eighteenth and nineteenth centuries provided both a novel and an extremely effective encouragement for the development of an *industrial* technology. In addition, the slowly opening political and social framework of late mercantilist society gave rise to social aspirations for which the new technology offered the best chance of realization. It was not only the steam-mill that gave us the industrial capitalist but the rising inventor-manufacturer who gave us the steam-mill.

The Expansion of Technology within the Market System Took on a New "Automatic" Aspect

Under the burgeoning market system not alone the initiation of technical improvement but its subsequent adoption and repercussion through the economy was largely governed by market considerations. As a result, both the rise and the proliferation of technology assumed the attributes of an impersonal diffuse "force" bearing on social and economic life. This was all the more pronounced because the political control needed to buffer its disruptive consequences was seriously inhibited by the prevailing laissez-faire ideology.

The Rise of Science Gave a New Impetus to Technology

The period of early capitalism roughly coincided with and provided a congenial setting for the development of an independent source of technological encouragement—the rise of the self-conscious activity of science. The steady expansion of scientific research, dedicated to the exploration of nature's secrets and to their harnessing for social use, provided an increasingly important stimulus for technological advance from the middle of the nineteenth century. Indeed, as the twentieth century has progressed, science has become a major historical force in its own right and is now the indispensable precondition for an effective technology.

It is for these reasons that technology takes on a special significance in the context of capitalism—or, for that matter, of a socialism based on maximizing production or minimizing costs. For in these societies, both the continuous appearance of technical advance and its diffusion throughout the society assume the attributes of autonomous process, "mysteriously" generated by society and thrust upon its members in a manner as indifferent as it is imperious. This is why, I think, the problem of technological determinism—of how machines make history—comes to us with such insistence despite the ease with which we can disprove its more extreme contentions.

Technological determinism is thus peculiarly a problem of a certain historical epoch—specifically that of high capitalism and low socialism—*in which the forces of technical change have been unleashed, but when the agencies for the control or guidance of technology are still rudimentary.*

The point has relevance for the future. The surrender of society to the free play of market forces is now on the wane, but its subservience to the impetus of the scientific ethos is on the rise. The prospect before us is assuredly that of an undiminished and very likely accelerated pace of technical change. From what we can foretell about the direction of this technological advance and the structural alterations it implies, the pressures in the future will be toward a society marked by a much greater degree of organization and deliberate control. What other political, social, and existential changes the age of the computer will also bring we do not know. What seems certain, however, is that the problem of technological determinism—that is, of the impact of machines on history—will remain germane until there is forged a degree of public control over technology far greater than anything that now exists.

Technological Determinism Revisited

Robert Heilbroner

"Do Machines Make History?" appeared in 1967 in *Technology and Culture.* Probably written a year earlier, it has now attained senescence by the standards of life expectancy of journal articles. I must confess that I had assumed it had long ago descended into the Great Limbo that awaits us all. All the greater pleasure, then, to see the article referred to as classic, although these days one is not certain if that refers to the standards of Cicero or Coca-Cola; more important, all the better reason to reassess its findings, now that they and I are both a quarter of a century older.

The article is an attempt to examine the idea of technological determinism as a powerful force of history—especially the history of large-scale socioeconomic transformations, of which the most important are the transition from feudalism to capitalism and the evolution of capitalism through its various stages. A great deal of attention is therefore devoted to the means by which "force" is generated in the flow of events through changes in the material basis of social life, and to the kinds of changes that this force effects. Hence, much of the original article is preoccupied with the problems of describing how technology evolves and with the linkages that connect it to social change.

These general themes still strike me as constituting the analytical core of the idea of technological determinism, and I shall accordingly devote a small part of this revisitation to a few emendations of and additions to my earlier conclusions. These changes are not of great significance—not because a rereading of my piece convinces me that there is nothing further of interest to be said on the topic, but simply because I cannot think of it. Perhaps for that reason, the aspect that today attracts me to the subject is one that 25 years ago had not yet caught my attention. It focuses on the question of what we take "history" to be, and why we are drawn to a technological mode of interpreting its palimpsest.[1]

That is only metaphor. The substantive problem is why we might be receptive to such a way of construing history. Here I must risk a generalization. It is that the attribute of "modern" historiography that most sharply distinguishes it from "premodern" (I use the terms to refer to styles, not to periods) is its treatment of the background panorama against which the foreground inquiry takes place. In premodern history, that panorama is dominated by inscrutable forces. In the foreground pharaohs and emperors come and go, city-states rise and fall, good kings follow bad and vice versa, but all the while, behind these adventures and misadventures we espy forces that obey a different causality—the whims of the gods, the dramas of cosmology and salvationism, or simply the intervention of chance, luck, and the like. "The only thing," writes Collingwood, "that a shrewd and critical Greek like Herodotus would say about the divine power that ordains the course of history is that . . . it rejoices in upsetting and disturbing things."[2]

In sharp contrast, the basic premise of modern historiography is that background forces arise from the same kinds of processes, and can be approached and apprehended at the same levels of understanding and explanation, as the objects of immediate scrutiny. This unification of foreground and background is perhaps most strikingly evident in the rise of an "economic" interpretation of history. From its initial eighteenth-century formulations (I think of Adam Ferguson's *Essay on Civil Society*) to Marxian materialism, Braudelian stratification, or neoclassical choice theory, the hallmark of modern historical work is an effort to establish filiations between the subject that has been singled out for treatment and the background against which the subject is displayed. Wars and political events, the staple

1. My dictionary (*Webster's New Twentieth Century*, 1971) tells me that a palimpsest is a "parchment or tablet that has been written upon or inscribed two or three times, the previous text or texts having been imperfectly erased and remaining, therefore, still visible." That is not a bad description of the writing of history. Technological determinism then becomes a prescription for grinding our lenses, enabling us to make out characters on the parchment that have not previously been seen or understood and thereby to write new texts or to read old ones in new ways.

2. R. G. Collingwood, *The Idea of History* (Oxford University Press, 1956), p. 22.

subjects of premodern history, are now of interest largely insofar as they embody and concretize background forces such as class struggle and rational maximizing. Indeed, it is characteristic of modern history that the "subject" becomes these very background forces themselves, and that individual figures or events are studied not so much for their intrinsic interest as to illustrate or instantiate the larger processes that interest the historian.

It is here, of course, that technological determinism enters the picture. Indeed, because modern historiography takes for granted that technology, like all background elements, must perforce penetrate the narratives of history-writing, the question changes from "Do machines make history?" to "How do machines make history?"—a change that opens the way for definitions, boundaries, and argument rather than for polemics or declarations of faith.

Technology and Material Life

Let us then look at some of the ways in which machines make history on the grand scale that interests us here. One such way comes immediately to mind: Machines make history by changing the material conditions of human existence. It is largely machines (here I use the term to denote both individual mechanisms and a general level of technological development) that define what it means to live in a certain epoch—at least, as an economic historian might define life. Elsewhere in this volume, Rosalind Williams points out that such an economics-oriented viewpoint not only begs serious questions but also establishes powerful agendas. I shall deal with that problem before I am done, but it is useful to begin by considering the intimate and pervasive engagement of machinery with everyday life.

A paradoxical aspect of this interconnection is that the engagement is, at the same time, the most immediately apparent example of how machines make history and the least satisfactory example of what we might mean by technological determinism. I shall spend only a few words on the first part of this assertion. If we wish to study a society unfamiliar to us, the best place to start is by grasping its material life. To understand the historical significance of Eileen Power's peasant Bodo, of Mantoux's Arkwright, or of Marx's Moneybags we must first become acquainted with the material circumstances of their lives. In the same way, an understanding of the events of contemporary history takes for granted a knowledge of the technological setting that shapes modern-day existence as the mountains and the sea shaped life in the premodern Mediterranean.

Yet such study does not answer the question of how machines make history. A recognition that the technological structure is inextricably entwined in the activities of any society does not shed light on the connection between changes in that structure and changes in the socioeconomic order.[3] Of course we would expect that the transition from feudalism to capitalism was profoundly connected with the rise of a new level of technological capability. But precisely what does "profoundly" mean? That is the question to which technological determinism promises to yield an elucidation, which is perhaps the closest thing to an answer that the question permits. Lacking such an elucidation, we cannot give any kind of *analytic* account of the manner in which changes in machines alter daily life. Hence, we may be tempted to depict the meaning of technological determinism as the ascription to machines of "powers" they do not have. This leads to impressive-sounding but ultimately unsupportable statements, such as Veblen's assertion that "the machine throws out anthropomorphic habits of thought" and Marx's expectation that the introduction of the railway into India would "dissolve" the caste system.[4]

The challenge, then, is to demonstrate that technology exerts its effects in generalizable ways. If technological determinism is to become a useful overlay for history's palimpsest, it must reveal a connection between "machinery" and "history" that displays lawlike properties—a force field, if we will, emanating from the technological background to impose order on human behavior in a manner analogous to that by which a magnet orders the behavior of particles sprinkled on a sheet of paper held above it or that by which gravitation orders the paths of celestial objects.

Is there such a force field? It is not difficult to find candidates for its order-bestowing task. Whitehead credited "routine," without which "civilization vanishes." Many have located the source of behavioral regularity in "human nature," variously described: Hume said that human motivation was a kind of repeating decimal in history.[5] For our purposes, what is lacking in these principles is an ability to translate the stimuli emitted by a changing technology into behavioral responses of a predictable kind with regard to transformations of a socioeconomic order. Routine will not serve that purpose; it tells us only that technological change will be resisted, not the form that resistance will take. Human nature fails for the same reason, even when it is reduced to its constitutive drives—aggression, self-preservation, or whatever—because we have no way of generalizing about the effects of technological change on these drives or about the effects of changes in the drives on the social framework.

What is needed, in other words, is a mechanism of a near-alchemical kind. A huge variety of stimuli, arising from alterations in the material background, must be translated into a few well-defined behavioral vectors. There must be a systematic reduction of complexity of cause into simplicity of effect, enabling us to explain how the development of new machineries of production can alter the social relationships constitutive of feudalism into those of capitalism, or those of one kind of capitalism into those of another kind. Such a mechanism seems impossible to imagine. What is perhaps even more imagination-defying is that it exists.

Economics as Force Field

The mechanism is, of course, economics, in the sense of a force field in which a principle of "maximizing" imposes order on behavior in a fashion comparable to the magnet and the gravitational pull of the sun. I shall consign to a note the meaning of the vexed term I have put into quotes. For our purposes, it will be sufficient to describe it in Adam Smith's straightforward formulation: "bettering our condi-

3. The recognition does, however, raise the important question of whether technology is itself a background or a foreground element. The answer depends on what we are seeking to investigate. If our interest lies in the sources of technological change itself, as in Joel Mokyr's book *The Lever of Riches* (Oxford University Press, 1990), technology becomes the foreground element on whose development impinge forces emanating from the socioeconomic background. If the dynamics of social formations themselves lie at the focal point of inquiry, the placements are reversed, and we seek causal factors in the technological background. It is the latter perspective that is normally associated with technological determinism.

4. Thorstein Veblen, *The Theory of Business Enterprise* (Scribners, 1932), p. 310. Marx from Michael Adas, *Machines as the Measure of Men: Science, Technology, and Ideologies of Western Dominance* (Cornell University Press, 1989), p. 240.

5. Alfred North Whitehead, *Adventures of Ideas* (Macmillan, 1933), p. 113; Hume, *An Enquiry Concerning Human Understanding*, section V, part 1.

tion [by] an augmentation of fortune."[6] This rough-and-ready description, for which "acquisitive mindset" is perhaps a more precise equivalent, is enough to reduce the varied stimuli of changes in the material environment to well-specified behavioral results.

At the risk of belaboring the obvious, economics accomplishes this remarkable feat by ignoring all effects of the changed environment except those that affect our maximizing possibilities. In this way, changes in technology, like changes in the weather or in our social situations, are depicted as loosenings or tightenings of constraints on our behavior, and these altered constraints are then perceived as changing our actions in sufficiently regularized ways to enable us to speak of "laws" at work in the marketplace or in the enterprise.

It is not necessary to discuss whether or not this regularized behavioral response can be considered a part of human nature.[7] It is enough that the acquisition of fortune becomes a widely noted "rule" of behavior in societies that leave behind the coordinating mechanisms of tradition and command for those of the market. Thus far in history these societies are, in fact, all members of the general social formation we call capitalism. It follows that economic determinism, and its technological correlate, have relevance only in the capitalist social order, in which, as Marx emphasized, the multifarious world of use values is transmuted into a one-dimensional world of exchange values.

In this social order, changes in the technological background are registered in changes in the price system, indicating the directions in which economic activity can most advantageously move and the forms it can most profitably assume. Thus the force field of maximizing allows us to elucidate how machines make history by showing the

6. Adam Smith, *The Wealth of Nations* (Modern Library, 1936), pp. 324–325. Maximizing described in terms of utility does not yield an action-directive capable of serving as an operational force field, insofar as utility maximization is tautologous with respect to behavior. Only a Smithian maximization of "fortune" provides the required specificity of response. Fortunately, this appears to be a reasonable description of economic life, most departures from the rule being self-correcting.

7. For support of this proposition see Jack Hirschleifer, "The Expanding Domain of Economics," *American Economic Review* (December 1985): 53, or Gary Becker, *The Economic Approach to Human Behavior* (University of Chicago Press, 1976). For a critique see chapters 1 and 7 of my own *Behind the Veil of Economics* (Norton, 1988).

mediating mechanism by which changes in technology are brought to bear on the organization of the social order. This leads in turn to the possibility of applying an analytic understanding to such large-scale social changes as the composition of the labor force and the hierarchical organization of work, not to mention the dynamics characteristic of economic activity as a whole.

Determinism as Heuristic

This immediately raises many questions and some specters. To begin, the triadic connection of technological determinism, economic determinism and capitalism does not mean that technology has no effects on noncapitalist society. The stirrup exerted a "catalytic" impact on the socioeconomic organization of Carolingian society, as did the spread of iron weapons from 1200 to 800 B.C. and the advent of printing in the fifteenth century. The difference is that precapitalist technological impingements do not affect their societies with the "logic" that comes only with capitalism's translation of use values into exchange values. The impact of precapitalist technical change therefore appears more contingent, less open to systematic elucidation, than when an economic force field guides its applications and consequences. We can say many more things about the "path" of technical change in the United States in the nineteenth century than about its course in ancient China or the Roman empire.[8] Perhaps there are other logics that would enable us to describe the interaction of technical change and social consequence with the degree of predictability and precision that is the very identifying characteristic of economics, but we do not know them.

Next, I hasten to add that in elevating economic determinism to the rank of a tutelary deity within capitalism, I am not asserting that economics is thereby entitled to assume the rank of the queen of the social sciences. I believe that what we call economic behavior is best understood as the sublimated expression of much deeper-rooted elements of "political" and "social" behavior—dominance and obedience—which can, in turn, be traced to the human experience of

8. Compare Ross Thompson, *The Path to Mechanized Shoe Production in the United States* (University of North Carolina, 1989), and chapters 2 and 9 of Mokyr, *The Lever of Riches.*

protracted childhood dependency.[9] Nonetheless, what is important is that "economic" behavior—that is, behavior motivated by the pursuit of exchange value—is set analytically apart from and above behavior motivated by other considerations, because it manifests a degree of orderliness absent from "political" or "sociological" activities. It may be that politics and sociology contain their own laws—I think of Alphonse Karr's "Plus ça change, plus c'est la même chose" and Lord Acton's pronouncements about the corrupting tendencies of power. Unlike economic behavior, however, these generalizations do not give us the translation mechanism of economics, enabling us—to take the present instance—to speak systematically about how machines will bring pressures to bear on socioeconomic formations.

Finally, technological determinism does not imply that human behavior must be deprived of its core of consciousness and responsibility. It avoids this trap insofar as it offers a heuristic of investigation, not a logic of decision-making. As such a heuristic, it presents a premise from which we can initially approach the interpretation of socioeconomic events, not an infamous "last instance" by which we are forced ultimately to resolve it. The premise is that living behavior is not random or chaotic, but marked by undeniable, although imprecise regularities: "undeniable" because predictable behavior is the basis on which all social life is raised; "imprecise" not only because there are obvious variances in behavior from one social entity to the next, but also because there exists a margin of behavioral indeterminacy within any given individual. Technological determinism then goes on to posit that the acquisitive mindset is a regular and dependable motive for behavior, at least in market-coordinated societies. Combining this persistent general drive and the margins of freedom characteristic of its concrete expression, we arrive at the formulation of a "soft determinism" which, however paradoxical to the philosopher, should present no great difficulties to the historian.

Degrees of Determinism

It is time to place these rather abstract considerations into more concrete perspective. Alfred Chandler's recent survey of American, British, and German capitalism during the period 1919–1948 provides a useful case in point.[10] Chandler's immediate interest is directed to explaining how the structures of capitalism were shaped by managerial styles that he describes as Competitive (American), Personal (British), and Cooperative (German). These styles, in turn, are shown to reflect underlying sociopolitical elements—what we might be tempted to call the respective "national characters" of the three nations. Chandler's important contribution is to point out how these differing responses to a common problem of industrial dynamics affected the developmental paths (and today, the development prospects) of the three nations.

This is an enlightening means of exploring the relation of technological and economic determinism, and the degrees of hardness and softness of these order-bestowing principles, although I hasten to add that these are my characterizations and not Chandler's. Chandler shows how the introduction of high-speed, continuous-run, capital-intensive machines led to a number of quite different, even mutually contradictory, institutional changes. Here is where technological determinism supplies many elucidations. The first is that the physical characteristics of the machinery of mass production can be discerned as the material cause of the phenomenon that Chandler is examining. These characteristics are not, however, the efficient cause. That lies in the translation of the engineering consequences of mass production into the economic stimuli of large changes in cost per unit of production—a translation that makes visible the force field of maximization to which activity is exposed in the market sphere of capitalism.

This is the "deterministic" aspect of the elucidation, and if that were all there were to it, we would expect that managements in all industrial capitalisms would respond in similar fashion. Now the equally important soft considerations enter, for the dramatic economies of scale and scope can be interpreted as opening new possibilities for—or new dangers from—corporate expansion. As Chandler shows, this makes way for a range of institutional responses, each a maximizing answer to the perceived economic situation. In this way, softness combines with hardness to throw analytic light on the different ways in which machines can influence the socioeconomic

9. Robert Heilbroner, *The Nature and Logic of Capitalism* (Norton, 1985), pp. 46–52.

10. Alfred D. Chandler, Jr., *Scale and Scope: The Dynamics of Industrial Capitalism* (Harvard University Press, 1990).

framework, even though all are obeying a common economic imperative.

Needless to say, this view of technological determinism is far removed from any kind of mechanical linkage. For instance, even where the specific characteristics of technology give rise to well-defined matrices of inter-industry connections, these linkages may not guide the course of economic growth if certain "soft" political and social preconditions are absent. Marx seems to have overlooked this possibility in speaking about the consequences of railroadization for India:

> . . . when you have once introduced machinery into the locomotion of a country, which possesses iron and coal, you are unable to withhold it from its fabrication. You cannot maintain a net of railways over an immense country without introducing all those industrial processes necessary to meet the immediate and current wants of railway locomotion and out of which there must grow the application of machinery to those branches of industry not immediately connected with railways. The railway system will become, in India, truly the forerunner of modern industry.[11]

Here Marx is talking about the railroad as an apparatus whose backward and forward linkages would eventually force a leap from one level of technological capability to another. Yet historical experience, not least that of India, has shown us that the logic of industrial linkage loses its peremptory force in determining the developmental path of a nation entangled in a preexisting global division of industry. The failure of centrally planned economies to duplicate the performance of capitalist economies is perhaps even more illustrative of such soft considerations. The history of Soviet industrialization has shown that technology is the servant and not the master of its associated system of sociopolitical directives, and that the use of more or less identical kinds of lathes, presses, or assembly-line configurations in centralized planning systems and in capitalist market systems will not result in anything like identical growth paths or performance levels.[12]

Indeed, if there is any substantial recantation that 25 years of observation has enjoined on me, it is to withdraw my earlier inclusion of "low socialism" along with high capitalism as a setting in which it would be possible to explain, even to foresee, the broad socioeco-

11. From Adas, *Machines as the Measure of Men,* p. 240.

12. See Nikolai Shmelev and Vladimir Popov, *The Turning Point: Revitalizing the Soviet Economy* (Doubleday, 1989).

nomic consequences flowing from changes in the machinery of production. I had assumed that low socialism meant a society that still embraced—and was still embraced by—capitalist-like motivations. In the absence of that premise, the application of technological—and, of course, of economic—determinism to the developmental tendencies of any form of socialism seems extremely uncertain.

The Bed of Procrustes

After so much has been taken away, what remains of technological determinism as a perspective from which we can consider how machines make history, always bearing in mind the socioeconomic meaning we attach to the last word in the question?

- Technological determinism gives us a framework of explication that ties together the background forces of our civilization, in which technology looms as an immense presence, with the foreground problem of the continuously evolving social order in which we live. The tie between the two is far from definitive, complete, or unambiguous, but it is the only such connection that we can make. As such, it has deep appeals to the modern temper, which recoils from introducing unknowability into the course of events. The deterministic view does not foreclose on a margin of indeterminacy (in some cases a very large margin), but its active search for regularities in, and lawlike aspects to, historical change remains the most powerful unifying capability we have. Whatever compression we may thereby suffer in our conceptions of ourselves as actors in history seems to me far less than would be felt if the very idea of a historic orderliness were shown to be utterly without basis. History as contingency is a prospect that is more than the human spirit can bear.
- Even in the most dramatic instances of technological determinism, as when we can trace the socioeconomic effects of the factory, the technique of mass production, or the modern-day computer, we can never eliminate the soft causal elements that are always present with, and within, those of the economic force field itself. Among these soft elements we must place many volitional elements, including most of what we call political decisions, social attitudes, cultural fads and fashions, and those aspects of maximizing itself in which the agent's final determination hinges on time horizons, risk aversion, and similar judgments about which no behavioral generalizations can be

made. Hence the clarifying power of determinism, even at its greatest, must always allow for some degree of uncertainty. This is perhaps only tantamount to saying that our conceptions of "history" cannot embrace either a fully determined or a wholly undetermined narrative of events—a state of affairs that no doubt reveals more about our psychological limitations than about the actualities of historical sequence, whatever they may be.

• Rosalind Williams protests that a deterministic heuristic, however soft, involves us in a cramping view of history's palimpsest, throwing aside much of its endlessly rich interpretational receptivity in favor of the thin stuff of behavioral "laws" and socioeconomic "formations."[13] I am very sympathetic to her denunciation of economism as a Procrustean approach to historical understanding, and no friend of the pretensions of economics to be a "universal science."[14] Disapproval is one thing, however; disavowal another. We live in a social order in which an economic calculus takes precedence over, and enters into the definitions of, many aspects of life. In the land of Procrustes, the standards of the innkeeper are those that apply. As long as economics constitutes the most powerful and pervasive motivational force, and the only one to which behavioral regularities can be ascribed, a perspective of soft determinism seems to me the one most likely to enable us to grasp the processes of history in which we are entangled.

Acknowledgement

I would like to thank Ross Thompson for his extremely valuable criticism.

13. These are my terms of derogation, but I think they catch her drift.

14. See my "Analysis and Vision in the History of Modern Economic Thought," *Journal of Economic Literature* (September 1990): 1111–1113, and "Economics as Ideology," in *Economics as Discourse,* ed. W. J. Samuels (Kluwer-Nijhoff, 1990).

Three Faces of Technological Determinism

Bruce Bimber

In this essay Bruce Bimber takes a different tack than Heilbroner. Specifically, he distinguishes three interpretations of technological determinism—the Normative, Nomological, and Unintended Consequences accounts—and develops a set of criteria with which to test the rigor and the legitimacy of these interpretations. Among other things, Bimber's strict approach "does not admit such distinctions as 'hard' and 'soft' determinism." He argues that only nomological accounts are "truly technologically deterministic." He tests his model by applying it to the work of Karl Marx and shows that Marx was not a technological determinist. "For Marx," he concludes, "technology was no more than one kind of important and efficient fuel for history's human engine."